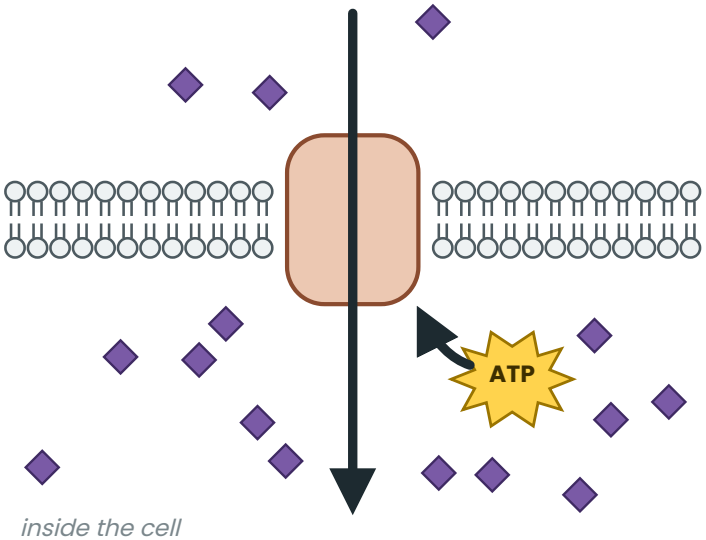
1

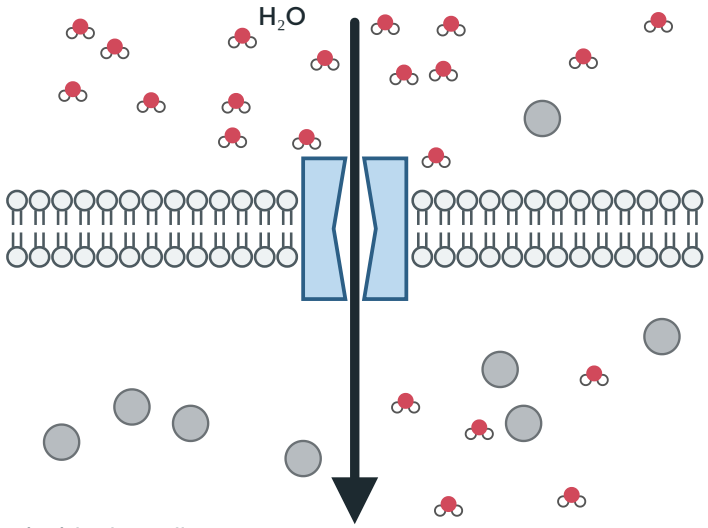
outside the cell



inside the cell

2

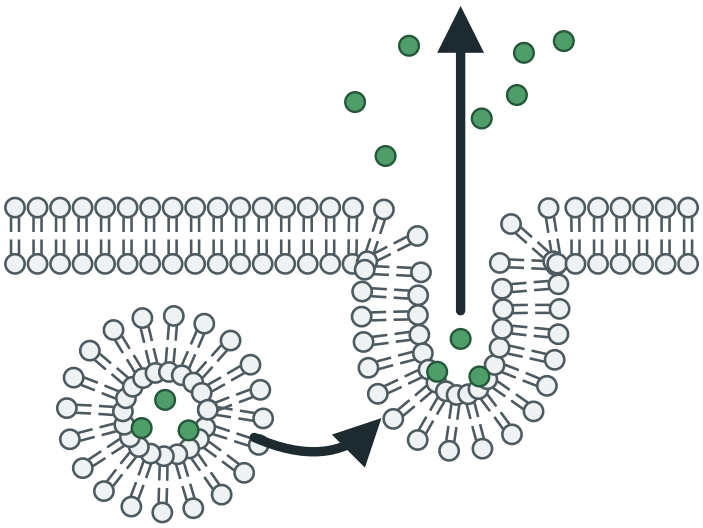
outside the cell



inside the cell

3

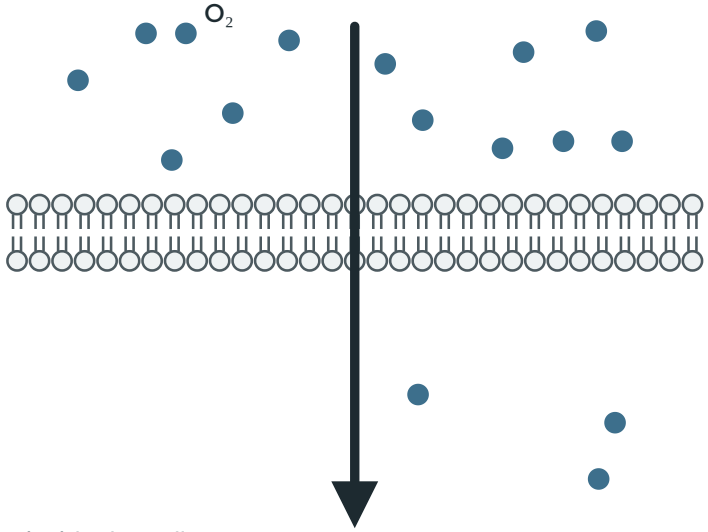
outside the cell



inside the cell

4

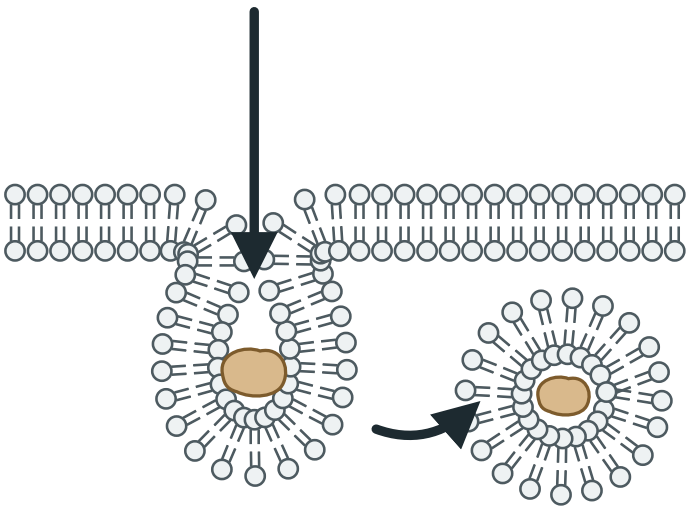
outside the cell



inside the cell

5

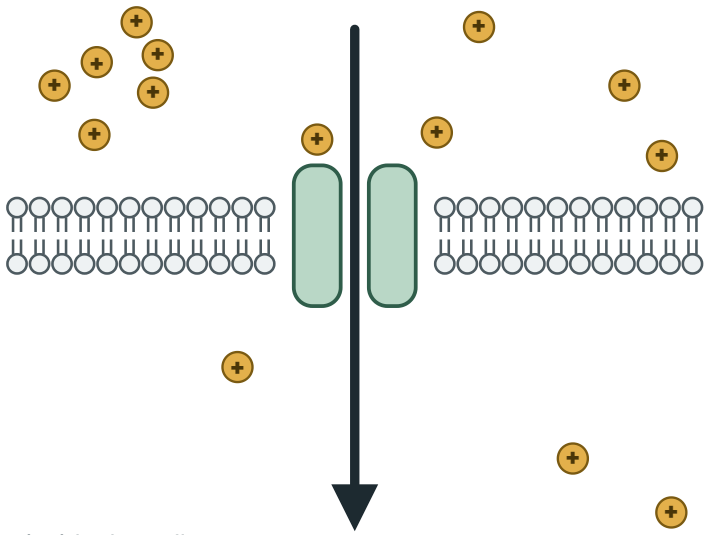
outside the cell



inside the cell

6

outside the cell



inside the cell

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A

Osmosis

B

Endocytosis

C

Simple diffusion

D

Facilitated diffusion

E

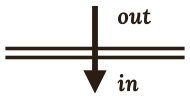
Exocytosis

F

Active transport

I

Spends ATP · active

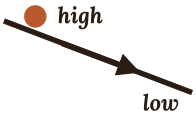


Direction. *Into the cell. The cargo is too large for any protein, so the gradient does not decide.*

How it crosses. *The membrane folds around the cargo and pinches off a vesicle.*

II

No ATP · passive

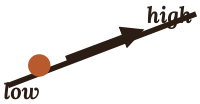


Direction. *Downhill: from high concentration to low.*

How it crosses. *Small molecules slip straight between the phospholipids. No protein is needed.*

III

Spends ATP · active



Direction. *Uphill: from low concentration to high, against the gradient.*

How it crosses. *A protein pump changes shape and pushes particles across.*

IV

Spends ATP · active

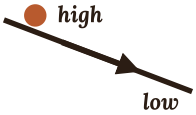


Direction. *Out of the cell. Large cargo, so the gradient does not decide.*

How it crosses. *A vesicle merges with the membrane and empties outside.*

V

No ATP · passive

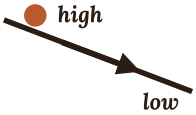


Direction. *Downhill: from high concentration to low.*

How it crosses. *A channel or carrier protein opens a path for a particle too big or too charged to pass the lipids.*

VI

No ATP · passive



Direction. *Downhill for water: from the side with more water to the side with more dissolved solute.*

How it crosses. *Only water crosses, through aquaporin proteins.*

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